

## HUBERT GAGADOSU

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1315 W Lincoln Hwy  
Unit 318, 60115  
DeKalb, IL, US.

### RESEARCH INTEREST

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Infectious Disease Modeling  
Mathematical Biology  
Ecological Models  
Biostatistics  
Mathematical and Computation Models  
Mathematical Optimization  
Bayesian Inference

### EDUCATION

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|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>PhD</b> | Northern Illinois University, DeKalb, IL, US.<br>PhD in Mathematical Science (Statistics)<br>(Main Courses: <i>Bayesian Statistics, Applied Regression Analysis, Discrete Models, Consulting, Longitudinal Data Analysis, Linear Model</i> ) | Present  |
| <b>MSc</b> | Sam Houston State University, Huntsville, TX, US.<br>MSc in Mathematics<br>(Main Courses: <i>Scientific Computations, Numerical Linear Algebra, PDF, Real Analysis, Topology, Algebraic Structures</i> )                                     | May 2025 |
| <b>MSc</b> | Nalanda University, Rajgir, Bihar, India<br>MSc in Ecology and Environment Studies<br>(Main Courses: <i>Remote Sensing &amp; GIS and its Applications, Green Accounting, Economics – Cost Benefit Analysis, EIA, Toxicology, Hydrology</i> ) | May 2021 |
| <b>BSc</b> | University of Education, Winneba, Ghana<br>Mathematics Education minor in ICT                                                                                                                                                                | May 2017 |

### RESEARCH EXPERIENCE

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| Graduate Student<br>Northern Illinois University, DeKalb, IL, US.<br>Topic: <i>A Mathematical Model of Infectious Colitis with an Epidemiological Active Medication Compartment: An SEIRM Framework</i> | Aug. 2026 |
| Graduate Student<br>Northern Illinois University, DeKalb, IL, US.<br>Topic: <i>Bayesian Analysis of Healthcare Expenditure Disparities in the USA by Region and Age</i>                                 | Aug. 2026 |

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| Graduate Student                                                                                                                                 |          |
| Sam Houston State University, Huntsville, TX, US.                                                                                                | May 2025 |
| Topic: <i>Simple Mathematical Model of Sudden Stratospheric Warming</i>                                                                          |          |
| Advisor: <i>Professor Edward Swim</i>                                                                                                            |          |
|                                                                                                                                                  |          |
| Graduate Student                                                                                                                                 |          |
| Nalanda University, Rajgir, Bihar, India                                                                                                         | May 2021 |
| Topic: Managing Plastic Waste of Rajgir: <i>Plastic to Liquid in Closed Burning</i>                                                              |          |
| Advisor: Dr Kishore Dhavala                                                                                                                      |          |
|                                                                                                                                                  |          |
| Undergraduate Student                                                                                                                            |          |
| University of Education, Winneba, Ghana                                                                                                          | May 2017 |
| Topic: <i>Using the number line to help Akatsi College of Education Students Overcome their difficulties in operations of negative integers.</i> |          |
| Advisor: <i>Dr. John Bijou Agbemaka</i>                                                                                                          |          |

## HONOURS AND AWARDS

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|-----------------------------------------------------|------------------|
| <b>Carol Feltz Scholarship Award</b>                |                  |
| Northern Illinois University                        |                  |
| Department of Statistics and Actuarial Science      | 2025 – 2026      |
|                                                     |                  |
| <b>Graduate Assistantship</b>                       |                  |
| Northern Illinois University                        |                  |
| Department of Mathematical and Statistical Sciences | 2026 – 2027      |
|                                                     |                  |
| <b>Graduate Assistantship</b>                       |                  |
| Northern Illinois University                        |                  |
| Department of Mathematics                           | Feb – May 2026   |
|                                                     |                  |
| <b>Graduate Assistantship</b>                       |                  |
| Northern Illinois University                        |                  |
| Department of Statistics and Actuarial Science      | Aug. – Dec. 2026 |
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| <b>Graduate Assistantship</b>                       |                  |
| Sam Houston State University                        |                  |
| Department of Mathematics                           | 2023 – 2024      |

## AFFILIATES

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American Statistical Association  
 American Mathematical Society  
 The National Society of Collegiate Scholars  
 International Epidemiological Association  
 All of US Research Program

## PROFESSIONAL UPGRADE

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|                                                                           |      |
|---------------------------------------------------------------------------|------|
| Introduction to Data Analysis                                             | 2023 |
| Excel Basics for Data Analysis                                            | 2023 |
| Data Visualization and Dashboards with Excel and Cognos                   | 2023 |
| Infectious Disease Modeling Using SIR/SEIR model and Developing SIR model | 2022 |
| Understanding Research Methods University of London                       | 2022 |
| Data Analysis in Python                                                   | 2022 |
| Data Analysis in R                                                        | 2020 |
| Data Analysis in Excel                                                    | 2020 |

## PUBLICATIONS

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- *A Mathematical Model of Infectious Colitis with an Epidemiological Active Medication Compartment: An SEIRM Framework* (Under Review)
- *Bayesian Analysis of Healthcare Expenditure Disparities in the USA by Region and Age* (Under Review)
- *Simple Mathematical Model of Sudden Stratospheric Warming* (Under Review)

## COLLEGE COURSE BOOKS WRITTEN/CONTRIBUTED TO (*Not Published*)

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- Trigonometry for College Students – Akatsi College of Education coursebook for students
- Further Algebra for College Students – Akatsi College of Education coursebook for students
- Probability and Statistics for College Students - Akatsi College of Education coursebook for students
- Euclidean Geometry and Trigonometry for College Students - Akatsi College of Education coursebook
- Calculus for College Students - Akatsi College of Education coursebook

## TEACHING EXPERIENCE

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**Northern Illinois University, IL, USA**

2025 - present

Graduate/Teaching Assistance, Department of Statistics

- Teach Undergraduate Courses (STAT 100, 200, 300, 400)
- Mark and record students' quizzes
- Have Office sessions to help resolve students' difficulties in mathematics and statistics

**Sam Houston State University, USA**

2023 - 2025

Graduate/Teaching Assistance, Department of Mathematics

- Taught Undergraduate Courses (Mathematics)
- Marked and recorded students' quizzes
- Held Office sessions to help resolve students' difficulties in mathematics

**Part-Time Assistant Lecturer, Department of Mathematics**

- Taught undergraduate students' different content, algebra, probability and statistics, geometry, trigonometry, and methods of teaching basic school mathematics

**University of Cape Coast College Distance Education, Ghana**

2016 – 2023

**Part-Time Assistant Lecturer, Department of Mathematics**

- Taught Bachelor students' mathematics; Trigonometry, Algebra, and Mathematics for SHS Teachers V (including limits, further differentiation and integration and their applications, experimental design, and hypothesis testing)

**Sogakofe Senior High School, Ghana**

Sept. 2018-July 2019

Ghana Education Service

**Full-Time Teacher, Department of Mathematics**

- Taught secondary school students' mathematics; Further Mathematics (main content are Calculus, Trigonometry, Probability and Statistics, Mechanics, Vectors, Algebra, and Matrices)
- Supervised students' examinations and always provide adequate feedback on their progress

**Akatsi College of Education, Ghana**

Aug. 2016 – Feb. 2017

**Lecturer, *Internship*, Department of Mathematics**

- Taught diploma students' mathematics; Trigonometry, Algebra, Further Algebra, Statistics and Probability, Calculus (differential and integral calculus), Business Mathematics, Methods of Teaching Basic school mathematics, and Geometry and Trigonometry
- Supervised students' project work

**Akatsi Senior High Secondary Technical School, Ghana**

May 2015 – Aug. 2015

**Class Teacher, *Attachment*, Department of Mathematics**

- Taught secondary school students' mathematics; Further Mathematics (main content are Calculus, Trigonometry, Probability and Statistics, Mechanics, Vectors, Algebra, and Matrices)
- Supervised students' examinations and always provide adequate feedback on their progress

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**PROFESSIONAL DEVELOPMENT TRAINING**

**Workshops**

- A Two Days Workshop on Aquifer Recharge Pit, A Collaborative Research of Nalanda University and ACIAR, Australia, 2020: I worked as an interviewer and interviewed the villagers' concerns about the project.
- A One-Day Workshop on Economic Dimensions Impact on Covid-19 Crisis – A Challenge in South Asian Countries, 2020: Was Organized by ARSD

College, Dhaulakuan, New Delhi – India.

- A-One Workshop on Day Gender and Social Security: Emerging Issues and Concerns in Post Covid Scenario – 2020: Was Organized by ARSD College, Dhaulakuan, New Delhi – India
- A One-Day Workshop on Sustainable Management of Health and Hygiene – Its Impacts on Post Covid-19 Environment – 2020: Organized by the Department of Management Studies, Gayatri Vidya Parishad College, India
- A One Day Workshop on Learn to Feel, Understand and Convey Emotions Right by National Institute of Disaster Management, Government of India - 2020

### **Courses with certificates**

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|--------------------------------------------------------------------------------|------|
| Certificate Program on Data Analysis in R Using STATCRAFT                      | 2020 |
| Three-Day R-Programming (Basics)                                               | 2020 |
| A Day Certificate Workshop on Linear Regression Model in Machine Learning      | 2020 |
| CII Young Professional Program – Climate Change & Environmental Sustainability | 2020 |
| CII Young Professional Program - Energy Efficiency                             | 2020 |
| A Series Certificate Program on Rainwater Harvesting                           | 2020 |
| US Institute of Peace - Conflict Analysis Micro Course                         | 2020 |

### **PROFESSIONAL DEVELOPMENT TRAINING**

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|---------------------------------------------------------------------------------------------|-------------|
| Colloquium Committee – Northern Illinois University, Department of Statistics 08/25 to date |             |
| Vice President, American Mathematics Society – Sam Houston State University Ch. 2023 – 2025 |             |
| Founder and Chairman, Akatsi Anyiheme Youth Network                                         | 2022 - date |
| Founder and Director, Generation Thinkers Network Foundation                                | 2019 – date |
| Chairman, EP Church, Ghana Akatsi Redeem Counseling Committee                               | 2018 – 2019 |
| Dep. Chairman, EP Church, Ghana Akatsi Redeem Welfare Committee                             | 2018 – 2019 |
| Chairman, National Service Personnel Association (NASPA), Akatsi                            | 2017 – 2018 |
| First Deputy Organizer, E.P Church, Ghana Student Union – UEW Chapter                       | 2015 – 2016 |

### **COMMUNITY SERVICE**

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| <b>Generation Thinkers Network Foundation</b>                                         | 2019 – date |
| Founder and Director                                                                  |             |
| A non-governmental organization committed to bringing societal change and development |             |
| <b>Founder: Akatsi Anyiheme Youth Network</b>                                         | 2022 - date |
| A community group committed to the development of the community                       |             |

### **LANGUAGES**

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**English:** Native Language, Distinguished levels in Listening, Speaking, Reading, and writing

## **OTHER SKILLS AND COMPETENCES**

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- R– developing the SIR model using R
- Statcraft – data analysis using Statcraft
- Excel – using Excel in data analysis
- Python/R Language
- Graphic Design
- Good English Communication Skills
- Team player

## **REFERENCES**

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To be provided when needed